

Sparse Self-Attention in Transformers

Cayley-Graph-Based Formalizations of Attention Masks

Semester Project Presentation

Aman Antil

Roll No.: 22CS30008

Department of Computer Science and Engineering
Indian Institute of Technology Kharagpur

Supervisor: Prof. Pabitra Mitra

B.Tech Semester Project

Outline

- 1 Motivation and Background
- 2 Cayley-Graph Viewpoint
- 3 Methodology and Results
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- Transformers are powerful because every token can interact with every other token.
- Standard self-attention therefore builds a dense $n \times n$ attention matrix.
- As sequence length grows, both compute and memory scale quadratically.
- Sparse attention is a natural direction for improving scalability.
- The key question is not only *how much* to sparsify, but *which links* to keep.

Problem Statement

Goal

Design sparse attention masks that reduce attention cost without severely harming downstream task performance.

- Different sparse structures preserve different information pathways.
- A purely ad hoc list of masks is difficult to compare systematically.
- We need a unified mathematical language for structured sparsity patterns.
- **Cayley graphs** provide a clean framework to formalize regular attention neighborhoods.

Dense Self-Attention Refresher

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

- Q , K , V are query, key, and value projections.
- Every token computes scores against every other token.
- The score matrix $QK^\top$ is dense in standard attention.

Cost

Attention matrix size = n^2

Compute $\sim O(n^2 d)$

Memory $\sim O(n^2)$

Why Sparse Attention?

- Fewer retained links means fewer attention scores to compute.
- Sparse masks can reduce both theoretical compute and memory.
- This matters more as context length increases.
- Sparse attention also injects an inductive bias into the model.

Key Tradeoff

Dense: $|E| = n^2$

Sparse: $|E| \approx nk, k \ll n$

Lower edge count improves efficiency, but quality depends on *topology*, not only on edge count.

Sparse Mask Families Studied

Mask	Core idea
Raw dense	Full token-to-token interaction; strongest baseline, highest cost
Window	Keep nearby tokens within a fixed local neighborhood
Dilation	Keep offsets at powers of two to create multiscale long-range hops
Circulant	Start from a window mask and add extra random connections in a repeated cyclic pattern
BigBird	Hybrid local + global + random sparse connectivity

- Backbone: **RoBERTa-base**
- Evaluation: **GLUE-style tasks** including SST-2, MRPC, RTE, CoLA, and STS-B

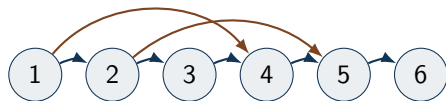
Cayley-Graph Viewpoint

From sparse masks to structured token graphs

Graph View of Attention

- Treat token positions as vertices.
- An allowed attention link from token i to token j becomes an edge.
- The sparse attention mask is therefore an adjacency pattern.
- Dense attention corresponds to a complete graph; sparse attention corresponds to a structured subgraph.

$$M_{ij} = 1 \iff (i, j) \in E$$



Local links +
selected long-range links

Cayley Graphs: Intuition and Definition

Lightweight group view

A group is a set with a composition rule. For sequence positions, the cyclic group $\mathbb{Z}_n$ is the most natural starting point.

$$\Gamma(G, S) : \quad V = G, \quad E = \{(g, gs) : g \in G, s \in S\}$$

- Vertices are group elements.
- A generator set S specifies the allowed moves.
- Edges connect each vertex to its translates by the generators.
- The same local rule is applied uniformly across the graph.

Why Cayley Graphs Are Useful Here

- They give a **generator-based description** of structured sparse masks.
- Uniform rules make masks easy to analyze and compare.
- Locality, periodic hops, and regular long-range reach can all be described through generator choices.
- Regularity is especially useful when the same sparsity template is repeated across many attention rows.
- The framework clarifies when a mask is translation-invariant and when it is not.

Design perspective

A mask family becomes a choice of graph topology rather than just a list of retained entries in a matrix.

Cyclic Group Viewpoint for Sequences

$$G = \mathbb{Z}_n, \quad S \subseteq \mathbb{Z}_n$$
$$j \in \mathcal{N}(i) \iff (j - i) \bmod n \in S$$

- Token positions are indexed modulo n .
- Generator offsets determine which relative positions are visible.
- Using a symmetric set $S = -S$ gives bidirectional-style neighborhoods.
- This is natural for masks defined by fixed offsets.

Interpretation

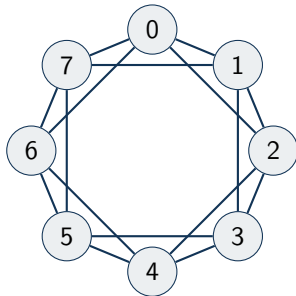
Choose the offset set once; then apply it to every token position.

- Good model for window, dilation, and circulant families.
- Less direct for masks with random or position-specific edges.

Window Attention as a Cayley Graph

$$S_{\text{win}} = \{\pm 1, \pm 2, \dots, \pm w\}$$

- Each token attends only to nearby positions.
- The graph emphasizes locality and short dependency paths.
- This matches tasks where nearby lexical or syntactic context is most informative.



Example: $S = \{\pm 1, \pm 2\}$

Dilation Masks as Structured Generators

$$S_{\text{dil}} = \{\pm 1, \pm 2, \pm 4, \dots, \pm 2^k\}$$

- Here the dilation mask uses offsets at powers of two rather than a single fixed stride.
- This creates a multiscale neighborhood: nearby access plus exponentially growing hops.
- The pattern is still structured and easy to describe through generators.
- It improves coverage across distance scales, but can still miss many task-critical intermediate alignments.

Takeaway

Powers-of-two dilation gives efficient multiscale reach, but its geometric spacing is still a strong inductive bias.

Circulant Masks as Cayley-Style Structured Patterns

- In this project, the circulant mask starts from a local window and adds a small set of extra random links.
- Those added links are encoded in a base pattern that is then repeated across rows by cyclic shifts.
- This keeps the implementation circulant, even though the extra links were initially chosen randomly.

$$M_{ij} = 1 \iff (j - i) \bmod n \in S_{\text{circ}}$$

- Under the Cayley-graph view, the realized mask is still described by the offset set S_{circ} .
- So randomness enters mask design, while the final adjacency remains a repeated structured pattern.

- BigBird combines **local**, **global**, and **random** connections.
- The goal is to balance locality with better graph diameter and broader reachability.
- It is popular because it offers strong sparsity while preserving some nonlocal access routes.
- Unlike pure Cayley-style masks, it is not fully translation-invariant: global tokens and random links break strict generator regularity.

BigBird philosophy

Local context
+ a few anchor tokens
+ random shortcuts

Contrast

Structured regular masks prioritize uniformity. BigBird prioritizes hybrid reachability.

Comparing the Masks Conceptually

Mask	Main inductive bias	Likely strength	Likely weakness
Raw	Full expressivity	Best coverage of interactions	Highest cost
Window	Strong locality	Stable for local syntax and sentiment cues	Limited long-range access
Dilation	Powers-of-two multiscale hops	Covers several distance scales efficiently	Geometric spacing can still miss key alignments
Circulant	Window + repeated random offsets	Adds nonlocal links while keeping row-wise structure	Randomly chosen links may not match task needs
BigBird	Local + global + random hybrid	Very high sparsity with nonlocal shortcuts	Random/global placement may still miss task-critical alignments

$$\rho = \frac{\text{number of retained attention links}}{\text{number of dense attention links}}$$
$$\rho = \frac{|\{(i, j) : M_{ij} = 1\}|}{n^2}$$

- Keep ratio gives a simple measure of how sparse a mask is.
- Smaller ρ means fewer retained token-to-token interactions.
- It is useful for comparing efficiency across mask families.
- But keep ratio alone is insufficient: two masks with similar density can behave very differently.
- Placement of edges matters as much as the number of edges.

Experimental Evaluation

RoBERTa-base under multiple sparse mask families

Experimental Setup

- Backbone model: **RoBERTa-base**
- Comparison variable: attention mask family
- Masks evaluated: raw dense, window, dilation, circulant, and BigBird-style sparse attention
- Task suite: **SST-2, MRPC, RTE, CoLA, STS-B**
- Focus of evaluation: relative quality retained under structured sparsity

Metrics

Accuracy for SST-2 and RTE, Accuracy/F1 for MRPC, Matthews correlation for CoLA, and Pearson/Spearman for STS-B.

Benchmark Overview

Task	What it measures	Why it matters here
SST-2	Sentiment classification	Mostly single-sentence semantics with strong local lexical cues
MRPC	Paraphrase detection	Requires fine-grained cross-sentence alignment
RTE	Textual entailment	Sensitive sentence-pair reasoning benchmark
CoLA	Grammatical acceptability	Strongly influenced by local syntactic patterns
STS-B	Semantic textual similarity	Requires rich sentence-pair interaction and score calibration

- This set spans both single-sentence and sentence-pair tasks.
- That makes it useful for testing how sparsity affects different interaction requirements.

Main Results Table (Part 1)

SST-2 Accuracy

Mask	Acc.
Raw	93.69
Window	93.34
Dilation	91.85
Circulant	92.43
BigBird	90.82

MRPC

Mask	Acc.	F1
Raw	88.48	91.62
Window	68.39	77.64
Dilation	70.83	79.3
Circulant	71.08	79.86
BigBird	70.09	79.53

Main Results Table (Part 2)

RTE Accuracy	
Mask	Acc.
Raw	75.09
Window	55.23
Dilation	52.71
Circulant	52.71
BigBird	52.71

STS-B		
Mask	Pearson	Spearman
Raw	0.9070	0.9051
Window	0.6731	0.6518
Dilation	0.562	0.5509
Circulant	0.6159	0.6034
BigBird	0.6201	0.6406

CoLA Matthews Correlation

Mask	Matthews
Raw	0.6109
Window	0.6209
Dilation	0.5829
Circulant	0.5599
BigBird	0.5905

Keep Ratio Table

Mask	ρ
Raw	1.00
Window	0.24
Dilation	0.17
Circulant	0.16
BigBird	0.09

Takeaway

Lower keep ratio gives higher sparsity, but it does *not* automatically give the best quality.

- BigBird is sparsest.
- Window keeps more links than the other sparse masks.
- Performance ordering does not follow keep ratio monotonically.

- **Raw dense attention is best overall** across the benchmark.
- **Window is the best sparse mask overall** in this project.
- **CoLA is the only clear case where sparse beats raw:** window 0.6209 vs raw 0.6109.
- Sentence-pair tasks degrade much more than single-sentence tasks.
- This suggests that sparse masks often damage the cross-token alignments required across paired inputs.

SST-2

- SST-2 is highly tolerant to sparsity.
- Raw 93.69 vs window 93.34 is a very small drop.
- Even aggressive sparsity keeps sentiment performance fairly strong.
- Likely reason: sentiment cues are often local and lexically concentrated.

CoLA

- Window 0.6209 slightly beats raw 0.6109.
- Local syntax appears especially important.
- A local mask may act as a useful regularizer.
- This is the clearest task where structured sparsity helps rather than hurts.

MRPC

- MRPC drops sharply under sparsity.
- Paraphrase detection needs richer cross-token interactions across two sentences.
- Circulant is the best sparse mask on MRPC.
- But even the best sparse score remains far below the raw baseline.

RTE

- RTE is extremely fragile under sparsity.
- Repeated 52.71 values suggest degenerate behavior or near-collapse.
- Window preserves slightly more signal than the other sparse masks.
- Entailment appears to require interaction patterns that fixed sparse masks do not preserve well.

STS-B and Semantic Similarity Inferences

- STS-B degrades heavily under sparsity, similar to the other sentence-pair tasks.
- Semantic similarity needs rich token-token interaction across both sentences.
- Window is the best sparse mask on **Pearson** (0.6731).
- BigBird is the best sparse mask on **Spearman** (0.6406).
- This suggests different masks may preserve different aspects of prediction quality:
 - absolute score calibration versus
 - ranking consistency.

BigBird vs Cayley-Style Structured Masks

- BigBird has the strongest sparsity among the tested masks: keep ratio 0.09.
- However, it is not empirically dominant on this benchmark.
- Aggressive sparsity alone is therefore insufficient.
- Global reachability is useful, but it is not enough by itself.
- Density and placement of edges must still match the task's interaction needs.
- In this study, regular local structure proved more reliable than stronger sparsity plus hybrid shortcuts.

Major Benchmark-Level Conclusions

- Full attention is the strongest overall.
- Sparse masks usually trade quality for efficiency.
- Window is the safest sparse approximation among the tested patterns.
- Topology matters more than tiny keep-ratio differences.
- There is no monotonic relation between keep ratio and performance.
- Task type strongly determines sparsity tolerance.
- Locality is the most reliable sparse prior among the tested masks.

Cayley-Graph Perspective and Overall Takeaway

- Cayley graphs provide a unified formal language for structured sparsity.
- Generator sets describe attention offsets cleanly and compactly.
- Window, dilation, and circulant masks can all be understood under this lens.
- This mathematical regularity is useful for **design**, **comparison**, and **interpretation**.
- But empirical suitability remains task-dependent.
- Graph-theoretic elegance helps formalize mask design; it does not replace downstream evaluation.

Conclusion and Future Work

Conclusion

- Sparse attention can work well, but performance depends strongly on task and mask structure.
- Window is the strongest sparse approximation in this project.
- CoLA benefits from locality.
- Sentence-pair semantic tasks suffer the most under fixed sparse masks.

Future Work

- Adaptive or learned generator sets
- Task-specific structured sparsity
- Longer-context evaluation
- Better hybrid local + nonlocal designs
- Hardware/kernel-level implementations
- Extending the Cayley-graph formalization to other backbones

Thank You